FLOODGAUGE · STEM CHECK

FloodGauge STEM check

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/floodgauge		
Source	docs/01b-stem-check.md @ dbb9a87		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	Initial check at the kit 1.9.1 upgrade, with the independent review; one detail failure fixed under the standing instruction (FLG-DDR-004 F1)

FloodGauge STEM check

Verdict: the working principles of all three concept options sit within known science. Of 32 claims, 15 are within known science and 17 are plausible and wait for a data sheet, a weighed mass or a TRL 4 test; none fails outright. The independent review (below) changed five verdicts and statuses. One detail failed on the first pass and was fixed under Amish's standing instruction of 2026-10-23: the solid ring floats of option C sank (FLG-DDR-004 F1). The check stays open until the plausible rows are closed. Method and rules: STANDARDS section 20. Calculations: `docs/04-calcs/stem_check.py`; the full sizing is in FLG-CAL-001.

Claims checked

Table 1. Claims, governing science and verdicts. S rows belong to option A (the current design), B rows to option B and C rows to option C. First-order checks use the geometry in `cad/src/model.py`; values marked "assumed" are listed under Assumptions below.

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S1	A: the street radar (60 GHz, module range 20 m) reaches the dry road 4.6 m below the lens and the water surface in between	Physics	Waves and radar	Free-space propagation of a 60 GHz pulse; module range 20 m (Acconeer product page, as cited in FLG-PRB-001, 2026-09-25)	5 m <= 20 m	Within known science	FLG-CAL-001 A1b
S2	A: street depth error stays within 10 mm when the radar module is accurate to 2 mm (assumed) and datum, pole expansion and ripple add 5, 2 and 3 mm	Physics	Measurement error	Root sum square of independent errors; the 2 mm module accuracy is assumed (S3), so the budget holds only if the module meets it	2 mm <= 7.88 mm: the module may be as poor as 7.9 mm before the 10 mm limit is reached, but pole bow in the sun (about 3 mm for a 5 K difference across the pole, independent review) and the worst-case sum of 12 mm are not inside the RSS	Plausible, to confirm at TRL 3	
S3	A: a 60 GHz carrier (wavelength 5.00 mm) gives a 2 mm range accuracy	Physics	Waves and radar	Coherent radar reads range from the phase of the echo: a 1 mm change in range shifts the phase by 4 pi / lambda = 2.52 rad, so 2 mm is several radians of phase	Phase argument supports mm-level accuracy, but the module's guaranteed accuracy, beam width through the PTFE lens and curb echo rejection are not in a data sheet yet	Plausible, to confirm at TRL 3	
S4	A: speed of sound c = 331.3 + 0.606 T (m/s) used	Physics	Acoustics	Ideal gas: c = sqrt(gamma R T / M) with gamma 1.4, R 287.05 J/	claimed 361.6 m/s at 50 C, estimate 360	Within known science	FLG-CAL-001 B2

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	to compensate the drain ranger over -20 to 50 C			(kg K); worst deviation 0.34 % over the range	m/s at 50 C (ratio 1.00)		
S5	A: drain level error stays within 20 mm at the 950 mm maximum range, with the NTC correcting air temperature to 2 K (10 mm module accuracy, 3 mm datum)	Physics	Acoustics	Two-way time of flight, c = 331.3 + 0.606 T; root sum square of module, temperature and datum errors (an uncompensated 10 K swing would add 16.8 mm); if the module accuracy is plus or minus (1 cm + 0.3 % of range) the error is about 13.6 mm, and humidity in a saturated basin adds a further 0.3 to 1 % to c, both to check on the data sheet	11 mm <= 20 mm	Within known science	FLG-CAL-001 C1
S6	A: the ultrasonic ranger (60 degree beam) reads the water surface 0.3 m to 1.25 m below it inside a 69 mm bore tube	Physics	Acoustics	The axial echo off the water is the shortest path for a smooth wall; but slot edges, joints, the cable loop and drops inside the tube can return an echo before the water echo, so the first echo is not always the water	Smooth-wall geometry supports it; echoes from the slot edges, the cable loop and condensation on the face are not excluded by the module (independent review), so the wet probe and a recorded empty-tube background are needed	Plausible, to confirm at TRL 3	
S7	A: the slotted stilling tube follows a basin rising at	Physics	Fluid mechanics	Orifice flow, v = Q / (Cd A) with Cd 0.6, head = v^2 / 2 g	0.198 mm <= 5 mm	Within known science	FLG-CAL-001 C4

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	10 mm/s with a lag head under 5 mm even with only the lowest slot row wet						
S8	A: the 40 x 40 x 1.6 mm aluminium arm survives a 50 kg person hanging on the head (misuse case)	Physics	Strength of materials	Cantilever to the top brace pin, sigma = M / Z; 6063-T6 class yield 214 MPa (typical, assumed)	60 MPa <= 214 MPa	Within known science	FLG-CAL-001 H3
S9	A: the 25 x 25 x 1.6 mm knee brace and its pins carry the same load	Physics	Strength of materials	L/r = 37 is below the Euler range, so Johnson inelastic buckling (28533 N) and pin bearing in 1.6 mm walls (11520 N) apply; the Euler value (73 kN) is not valid	1504 N <= 1.152e+04 N	Within known science	FLG-CAL-001 H3
S10	A: two band clamps alone hold the arm against the twist of a 35 m/s wind along the street (60 mm pole; on a 40 mm pole friction gives 24.0 N m)	Physics	Strength of materials	Coulomb friction, mu 0.30, 1 kN preload per band (FieldNode assumption); wind drag, q = 0.5 rho v^2	24.8 N m <= 36 N m	Plausible, to confirm at TRL 3	S11 carries the twist as a second path; on a 40 mm pole it is the only one
S11	A: the M8 A4-80 anti-rotation through-bolt holds the arm against the same twist	Physics	Strength of materials	The bolt bends as a 21.7 mm cantilever out of the pole wall through the spacer (independent review corrected the first couple model): M = f_y Z,	24.8 N m <= 44.6 N m	Plausible, to confirm at TRL 3	FLG-CAL-001 H2b; FLG-DDR-004 F2 (class raised from A4-70 to A4-80)

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				f_y 600 MPa (A4-80), stress-area section; plate load 52 mm from the pole axis			
S12	A: the added mass on the pole stays at or under 5 kg	Engineering and technology	Mass budget	Sum of part volumes times density from the CAD model plus FieldNode 2.45 kg and cables (FLG-CAL-001 I1, I2)	4.994 kg <= 5 kg	Plausible, to confirm at TRL 3	weigh at TRL 4; FLG-DEC-001 item 3 (bracket plate to 2.5 mm)
S13	A: pole expansion moves the lens by about 2 mm over a 35 K swing	Physics	Thermodynamics	Linear expansion, dL = alpha L dT with steel alpha 12e-6 per K	claimed 2 mm, estimate 1.98 mm (ratio 1.01)	Within known science	FLG-CAL-001 B1
S14	A: stored energy runs the sensors for 5 days without sun at event sampling	Physics	Energy balance	Usable 15.36 Wh (FND-CAL-001), draw from the sampling cycle energy of FLG-CAL-001 D1 to D4; days at -20 C used	5 days <= 47 days	Within known science	FLG-CAL-001 D4
S15	A: normal-mode uplinks (every 15 min, 20-byte payload, SF9) stay within The Things Network fair use of 30 s airtime per day	Engineering and technology	Radio	LoRa time on air (Semtech formula); fair use 30 s/day (The Things Network, checked 2026-10-31)	23.7 s/day <= 30 s/day	Plausible, to confirm at TRL 3	FLG-CAL-001 E1; PR-007 stays at risk at SF10 and slower
S16	A: event-mode uplinks every 60 s at SF9 stay within the EU868 1 % duty cycle	Engineering and technology	Radio	Duty cycle = time on air / interval; limit 1 % (EU868 sub-band, ETSI EN 300 220; text not reopened, see FLG-REG-001)	0.41 % <= 1 %	Within known science	FLG-CAL-001 E1
S17	A: the drain head face, 0.9	Physics	Fluid mechanics	Hydrostatic head p = rho g h; IP67 is 1	0.9 m <= 1 m	Plausible, to	PR-009 stays at risk: 72 h

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	m under water in the worst design flood, stays inside the 1 m rating of its IP67 module			m for 30 min (IEC 60529)		confirm at TRL 3	duration unverified (FLG-CAL-001 K1)
S18	A: the drain head survives 72 h under silty stormwater at 2 m (IP68)	Engineering and technology	Ingress protection	IP68 is defined by agreement with the maker, not by the code; the potted electronics are sealed but the transducer face keeps the stock IP67 seal	Not shown by calculation; needs the module maker's rating or an immersion test at TRL 4. Pressure at 2 m is 19.6 kPa	Plausible, to confirm at TRL 3	
S19	A: stainless A4 fasteners in a 6063 aluminium arm and bracket survive road-salt spray for the life of the pilot	Chemistry	Corrosion	Galvanic couple of stainless steel and aluminium; the cathode (fastener) is small and the anode (aluminium) large, the favourable area ratio, and anodising or sleeves separate them	Favourable area ratio and crush sleeves; chloride splash can still pit the aluminium at the contact, to confirm with a coating or an isolating washer at TRL 3	Plausible, to confirm at TRL 3	
S20	A: the PVC stilling tube stays below its service temperature in a shaded basin	Chemistry	Materials	PVC-U typical maximum continuous service temperature 60 C (manufacturer data, to confirm for the chosen pipe); PVC-U also turns brittle in the cold, so impacts at -20 C are a handling risk	35 C <= 60 C	Within known science	BOM item 5
S21	A: 5 x 50 mm slots in the stilling tube stay open in silty, leaf-	Biology	Fouling	Slots 5 mm wide pass silt but clog with leaf litter and biofilm over weeks in warm standing	Open area is 10000 mm2 against a 3739 mm2 bore, so a quarter	Plausible, to confirm at TRL 3	

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	laden stormwater for a season			water; 40 slots in 10 rows leave 75 % of the open area if a quarter clog	clogged still passes the flow of S7; cleaning interval to assess		
S22	A: an event-mode alert reaches the service within 120 s at SF10 on a private gateway, worst case without a lost packet (10 s sample, two confirming samples, wake, airtime, duty-cycle wait, 4.7 s network)	Mathematics	Timing budget	Sum of the delays; SF11 gives about 134 s and SF12 about 217 s, so PR-006 holds to SF10 only (independent review)	81 s <= 120 s	Within known science	FLG-CAL-001 F4
S23	A: the street head's I2C bus stays within 400 pF on a 3 m cable	Engineering and technology	Electrical interface	Bus capacitance = cable length x pF/m + 50 pF: 350 pF at an assumed 100 pF/m; a 150 pF/m cable gives 500 pF	Passes at the assumed cable; the real cable's capacitance is on its data sheet	Plausible, to confirm at TRL 3	
B1	B: a 0 to 20 kPa vented transducer covers 600 mm in the street (5.9 kPa) and the 1.25 m basin range (12.3 kPa)	Physics	Fluid mechanics	Hydrostatic pressure p = rho g h, gauge pressure against the vented atmosphere	12.26 kPa <= 20 kPa	Within known science	FLG-OPT-001 Table 1
B2	B: street depth error stays within 10 mm (0.25 % of a 20 kPa range, 0 to 40 C water	Physics	Measurement error	Root sum square; water density 999.8 to 992.2 kg/m3; silt density rise = SSC (1 - rho_w / rho_s), rho_s 2.65;	8.7 mm <= 10 mm	Plausible, to confirm at TRL 3	FLG-OPT-001 Table 2

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	density, 5 g/l silt, 5 mm datum)			accuracy class assumed			
B3	B: the vented cable keeps the atmospheric reference dry for years	Chemistry	Humidity and condensation	A vented gauge transducer breathes through its cable; condensation or water in the vent tube shifts the zero	Desiccant or a PTFE vent membrane is the usual remedy; lifetime in a flooded basin is not shown	Plausible, to confirm at TRL 3	
B4	B: a stand pipe on the curb face does not split or crush the transducer when street water freezes at -20 C	Physics	Phase change	Water expands about 9 % on freezing and can split a closed pipe; the bottom cap has four drain holes, so only a film freezes	Drain holes leave no standing column after an event; a trapped film is expected to be harmless, to confirm	Plausible, to confirm at TRL 3	
C1	C: the street float (hollow shell, 42 mm OD) floats with its magnet at under 70 % of its volume submerged	Physics	Fluid mechanics	Archimedes: weight (shell 29.2 g, plus about 3 g of foam) = displaced water; solid ring first pass was 44.4 g against 33.7 g of buoyancy and sank; the voids are filled with closed-cell foam (about 3 g) so a leaking shell still floats (FLG-DDR-004 F1)	61 % of buoyancy used <= 70 % of buoyancy used	Within known science	FLG-DDR-004 F1 (fix rule); model options_bc.py
C2	C: the basin float (hollow shell, 56 mm OD) floats with its magnet at under 70 % of its volume submerged	Physics	Fluid mechanics	Archimedes: weight (shell 47.0 g, plus about 3 g of foam) = displaced water; solid ring first pass was 106.5 g against 97.4 g and sank	48 % of buoyancy used <= 70 % of buoyancy used	Within known science	FLG-DDR-004 F1 (fix rule); model options_bc.py
C3	C: a reed chain at 25 mm pitch	Mathematics	Quantisation	Resolution of a switch chain = pitch; the level is	True as stated. It exceeds the 10 mm street		

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	reads the level to 25 mm steps (plus or minus 12.5 mm)			the highest closed reed, with one or two reeds closing at once, so the error is at most one pitch	target of PR-002, so this is scored against the user requirements in FLG-OPT-001, not a failure of the science	Within known science	
C4	C: a ring magnet beside the sealed stem closes a glass reed switch 1.9 mm from the stem axis	Physics	Magnetism	Reed switches close at 10 to 30 ampere-turns of applied field; a sintered ring magnet at a few millimetres gives a field far above that; hysteresis spreads the closing height by a few mm	Plausible; the closing height and its spread need the reed's data sheet, assumed typical 10 to 30 AT	Plausible, to confirm at TRL 3	
C5	C: the floats move freely on 16 mm stems in silty stormwater and at -20 C	Biology	Fouling	Silt and biofilm in the 0.9 mm radial gap between bore and stem can bind a float; ice binds it solid in a frozen tube	Plausible; the bottom cap drain holes empty the street pipe, the basin float is not protected from ice, to assess in OPT scoring	Plausible, to confirm at TRL 3	

Verdicts: **Within known science**, **Plausible, to confirm at TRL 3**, **Fails, fix proposed**, **Fails**.

Assumptions

- Radar module accuracy 2 mm, 25 degree beam and range 20 m (Acconeer product page, as cited in FLG-PRB-001, 2026-09-25): accuracy and beam are assumed, there is no data sheet in the repo yet.
- Aluminium 6063-T6 class, yield 214 MPa, E = 69 GPa; band clamp preload 1,000 N and friction 0.30 (FieldNode assumption); pole wall 2.5 mm and f_u 360 MPa.
- Pressure transducer class 0.25 % of a 20 kPa full scale for option B; typical, not quoted.
- Reed switch operate range 10 to 30 ampere-turns for option C; typical, not quoted.
- Polypropylene 900 kg/m3 and sintered NdFeB 7,500 kg/m3 for the floats.

Fields not relevant

- **Biology:** no living material; the only biological claims are fouling of the slots and floats (S21, C5).
- **Chemistry:** no reaction or dosing; the claims are corrosion and polymer service temperature (S19, S20) and vent condensation (B3).

Concept options

- **A, radar and ultrasonic (current design):** within known science; S2, S3, S6, S10 to S12, S15, S17 to S19, S21 and S23 are plausible and wait for the radar and ranger data sheets, the weighed mass, the module immersion rating and a coating decision.
- **B, hydrostatic pressure:** within known science; the 8.7 mm street error (B2) is inside 10 mm only with the assumed transducer class, and the vent and freezing rows (B3, B4) are plausible.
- **C, float and reed chain:** within known science after fix F1; the 25 mm pitch (C3) is a requirement miss for PR-002, not a physics failure, and is scored in FLG-OPT-001.

Fix made under the standing instruction

F1, hollow floats (FLG-DDR-004). The first model of option C used solid polypropylene ring floats with an 8 mm ring magnet. Archimedes' principle showed they sink: the street float weighs 44.4 g against 33.7 g of buoyancy, the basin float 106.5 g against 97.4 g (`stem_check.py` prints the record). The fix keeps the working principle: sealed hollow polypropylene shells with 1.6 mm walls and a 5 mm ring magnet, the street float widened from 38 to 42 mm. They then weigh 29.2 g and 47.0 g and ride at 61 % and 48 % of their buoyancy, so C1 and C2 close. The independent review added closed-cell foam in the voids so that a leaking shell still floats. The model, `cad/fidelity.json` and FLG-DEC-001 carry the change. A second fix, F2 (through-bolt class), came from the review (FLG-DDR-004).

Independent review

A second agent (Claude, model claude-opus-5-5, run headless with no tools beyond reading the files and arithmetic) saw only this STEM check, the calculation note FLG-CAL-001, the product requirements FLG-PRS-001 and `sizing.py`, not the reasoning behind them. It was asked to refute the headline results and every "Within known science" verdict. It could not run `sizing.py` (the model is not in its folder) and recomputed from the stated inputs. Most results survived: the error budgets, airtime, energy, tube lag, arm stress, the speed-of-sound formula and the cost arithmetic. It found five things that changed a verdict or a requirement status, and ten smaller points.

Changed a verdict or status, and what was done (fix rule, FLG-DDR-004):

1. **S11, the through-bolt (refuted).** The couple model across the full pole was the wrong load path; the bolt bends as a cantilever through the spacer and resists 28 to 48 N m, not 922 N m. The factor of 37 was wrong. Fix F2: class A4-80 bolt, bending model in CAL H2b, factor 1.8 (1.3 for A4-70); S11 is now "Plausible, to confirm at TRL 3" and PR-015 is restated. Small poles remain thin: friction alone gives 0.97 on a 40 mm pole.
2. **S12 and PR-015, mass.** A 6 g margin on 4.994 kg is inside the accuracy of the estimate, and the 2.5 mm plate fallback saves only 45 g. PR-015 is now "At risk" (it was "Met on paper").
3. **PR-006, latency.** Held only to SF10; in event mode the 95th percentile is 123 s at SF11 and 201 s at SF12 (reproduced in CAL F4). PR-006 is limited to SF9 and SF10; new row S22.

4. **S2, street depth error.** It rested on the assumed 2 mm radar accuracy, which S3 calls plausible. S2 is now "Plausible, to confirm at TRL 3".
5. **C1 and C2, floats.** The sealed hollow shell moves the failure to leaks and pressure cycling of the sealed air. Fix F1 now fills the voids with closed-cell foam, so a leaking shell still floats; the model carries the foam as its own part. The rows stay within known science for buoyancy; sealing and creep are for the TRL 4 leak test.

Smaller points, applied: the brace used the Euler load where L/r = 37 makes it invalid (Johnson buckling and pin bearing now used: capacity 11.5 kN against 1.5 kN, S9); the wall-echo reasoning in S6 was wrong for slot edges and the cable loop (text corrected); the drain error if the ranger is plus or minus (1 cm + 0.3 % of range) and humidity in a saturated basin are noted in S5 and make the NTC mandatory; PVC brittleness at -20 C is noted in S20; I2C capacitance at 150 pF/m is now S23; the pole bow term (about 3 mm for 5 K across the pole) is noted in S2; stale text about the 150 mm band in CAL F was corrected.

Not changed: the float dead band (about 26 mm of water before a 42 mm float lifts) and the false-reading step rule against surges faster than 300 mm/min are logged for the design review; the sun-induced pole bow is not yet in the error budget. The reviewer also asked whether the 214 MPa yield is typical rather than minimum; with a 170 MPa minimum the arm still passes at a factor of 1.4.

Sources

- Acconeer. "Products." https://www.acconeer.com/products/. Range 20 m, as cited in FLG-PRB-001 and first checked 2026-09-25; not reopened in this session.
- DFRobot. "A02YYUW waterproof ultrasonic sensor." https://www.dfrobot.com/product-1935.html. Checked 2026-09-25 (3 to 450 cm, plus or minus 1 cm, IP67).
- The Things Network. "Fair Use Policy explained." https://www.thethingsnetwork.org/forum/t/fair-use-policy-explained/1300. Checked 2026-10-31 (30 s/day).
- Legal Information Institute. "47 CFR 15.255." https://www.law.cornell.edu/cfr/text/47/15.255. Checked 2026-10-31.
- Archimedes' principle, hydrostatic pressure p = rho g h, Euler buckling, Coulomb friction, ideal-gas speed of sound and orifice flow: standard textbook relations.
- IEC 60529 (IP codes) and EU868 duty cycle rules are relied on from general knowledge and not reopened in this session (FLG-REG-001).
- FLG-CAL-001 (docs/04-calcs/01-sizing.md) and FLG-PRS-001 for the values the rows close against.